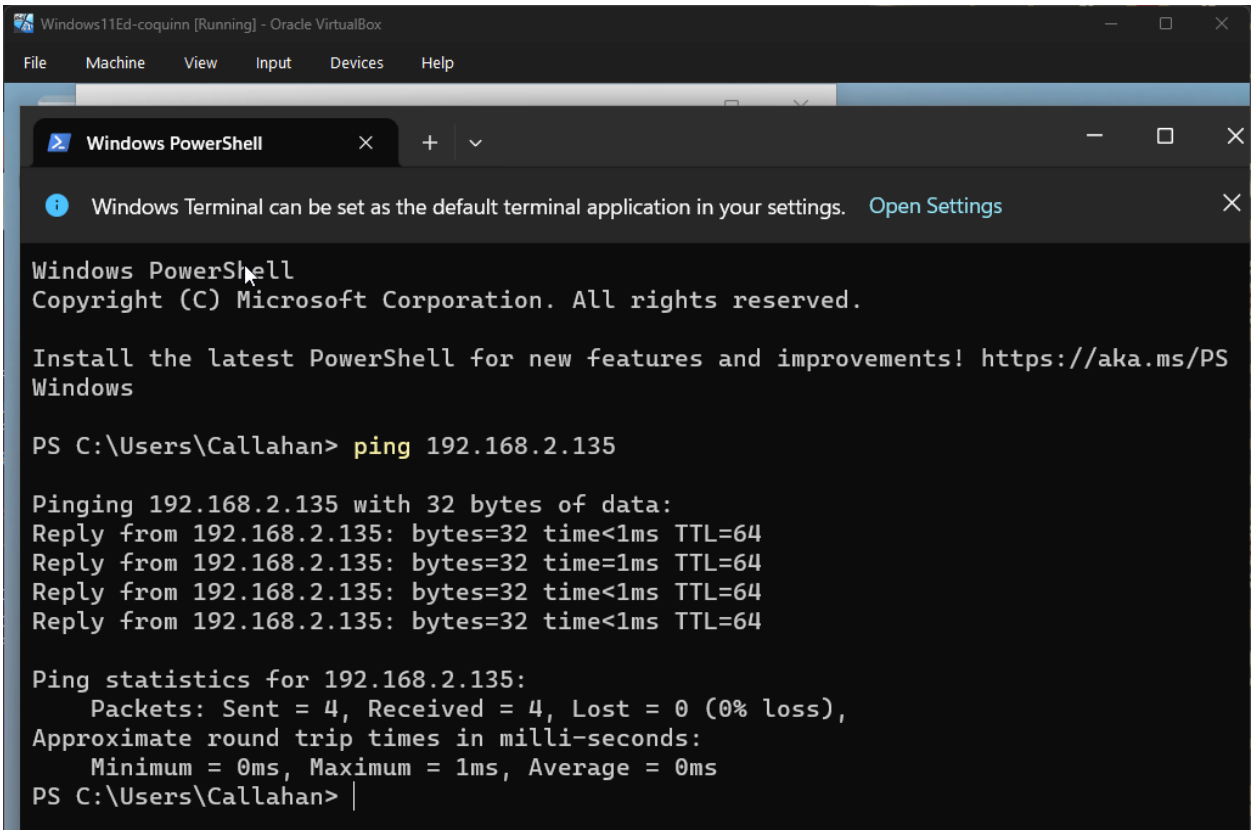
Network Segmentation & Containment

ISEC3078 – Lab 5

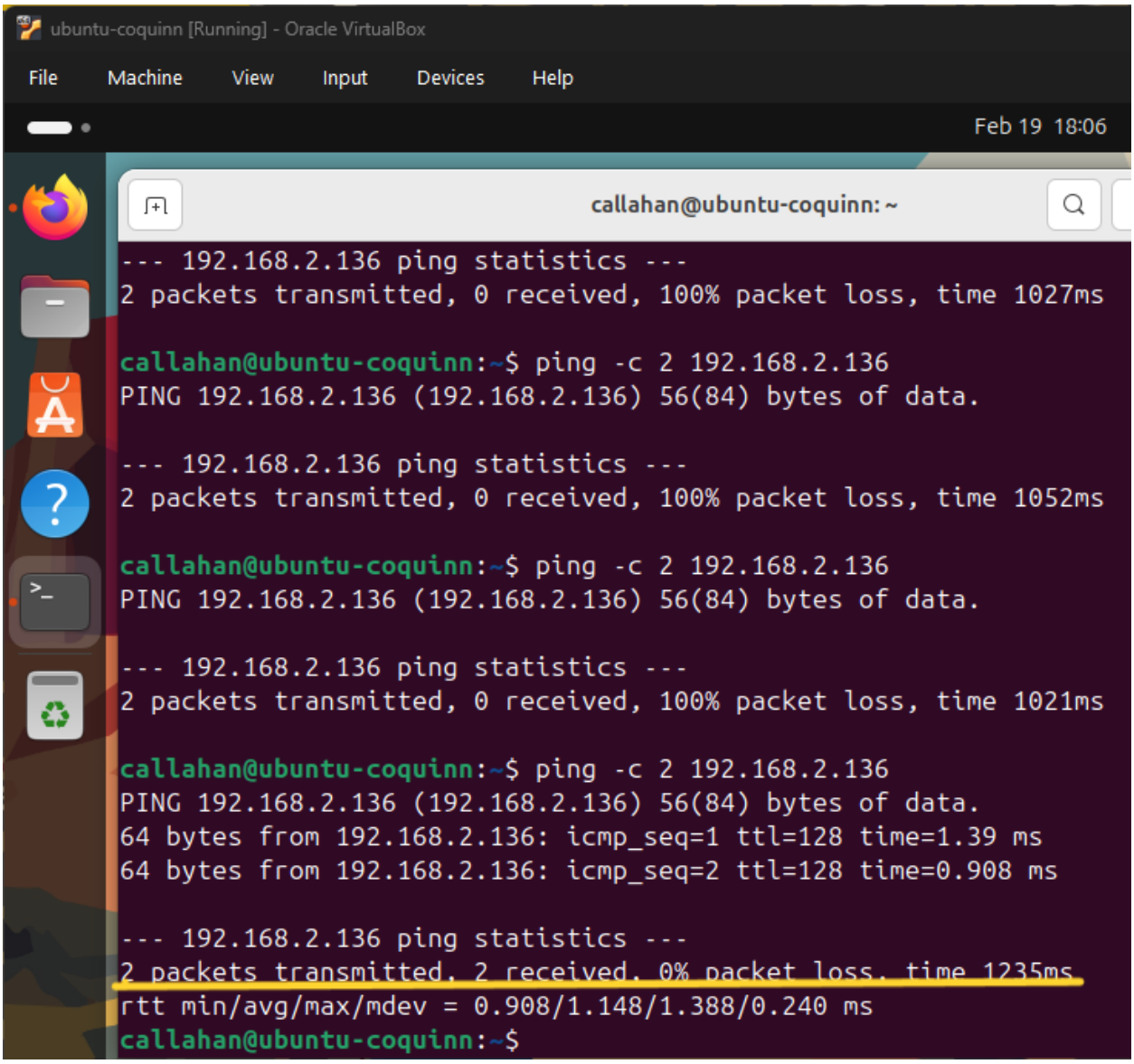
Callahan O'Quinn

1 VERIFY A FLAT NETWORK

1.1 PING FROM WINDOWS VM TO UBUNTU VM



1.2 PING FROM UBUNTU VM TO WINDOWS VM

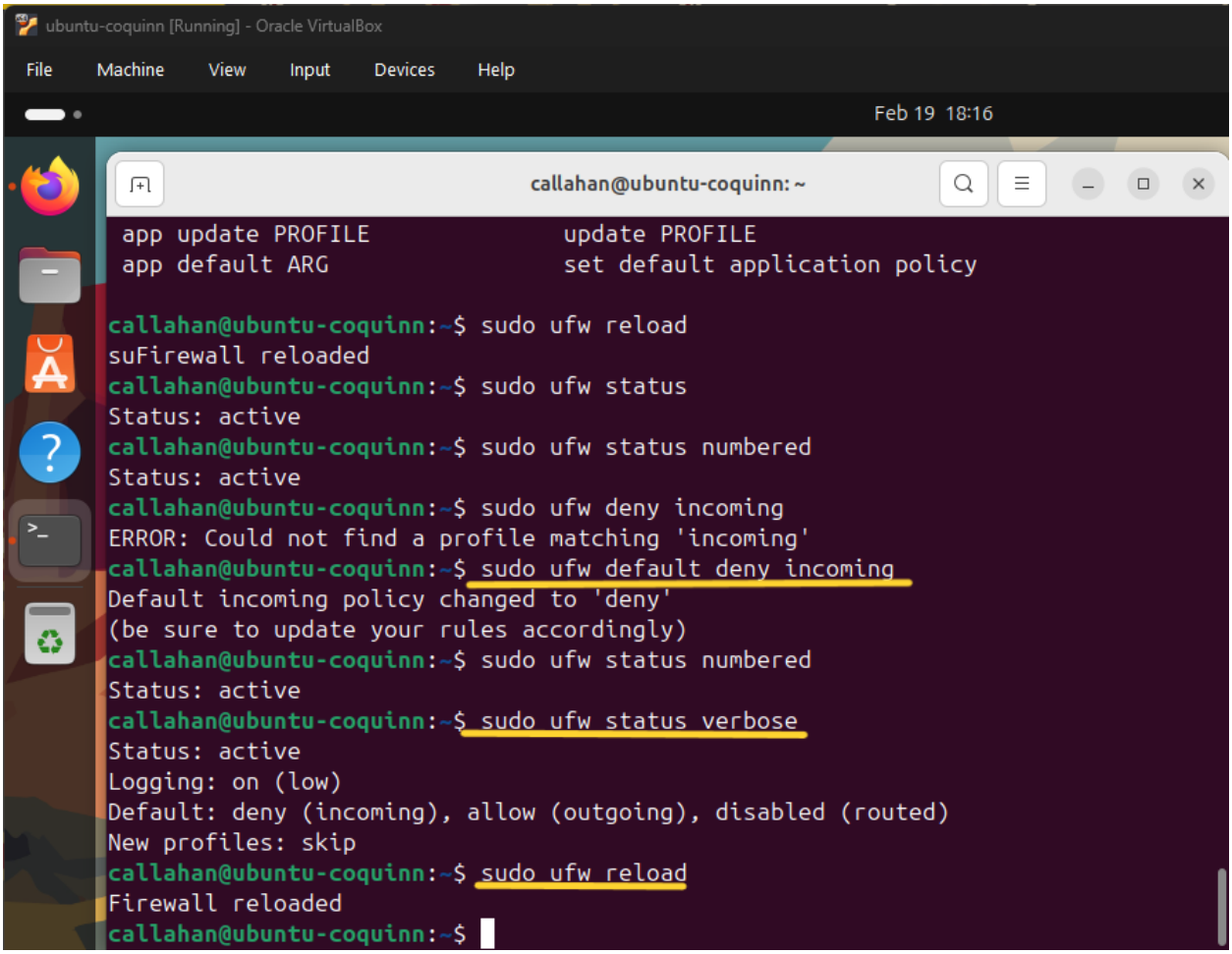


1.3 QUESTIONS

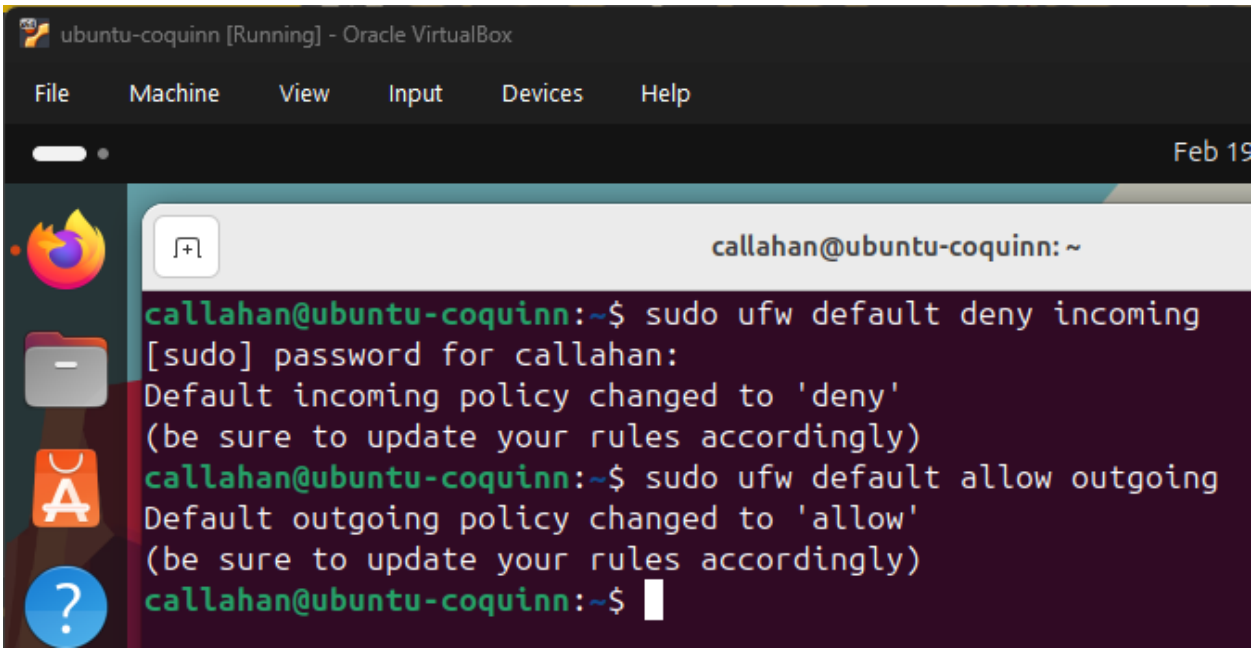
- Why is this considered a flat network?
  - It is a flat network because there is no segmentation between devices, meaning that communication does not need to be routed before reaching its destination.
- What risks does this create during an attack?
  - It is easy for attackers to move laterally through a flat network, because it has less boundaries between devices. Ransomware for example can easily hop to other devices on a flat network.

2 SIMULATE A FIREWALL BETWEEN NETWORK SEGMENTS

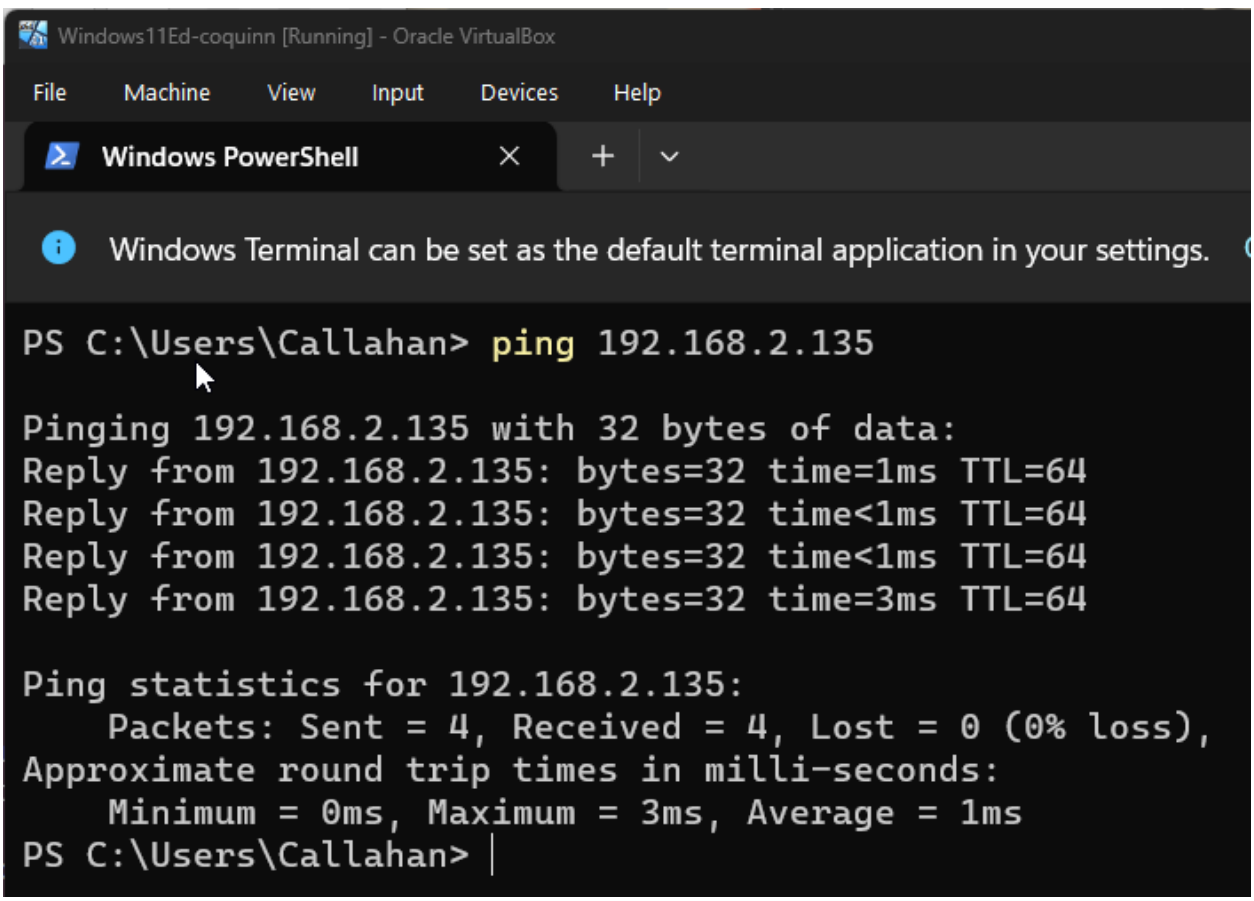
2.1 ENABLE THE FIREWALL



2.2 SET DEFAULT FIREWALL RULES



2.3 TESTING CONNECTIVITY

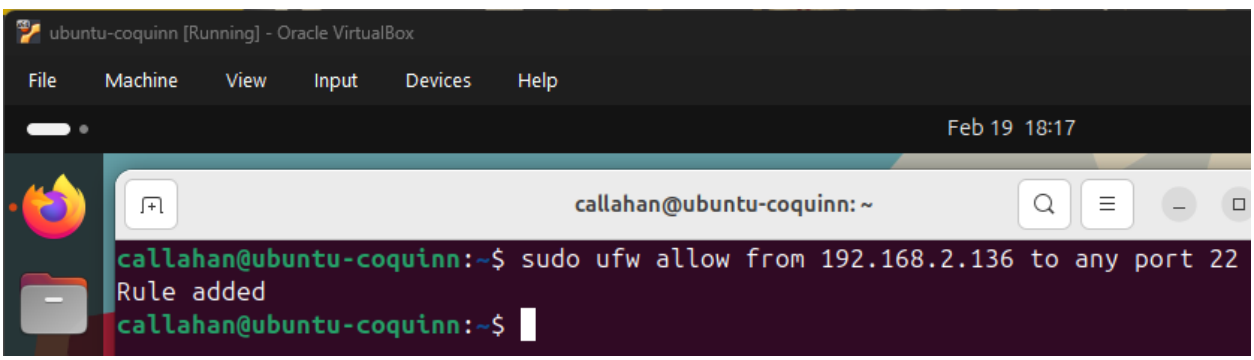


2.4 Questions

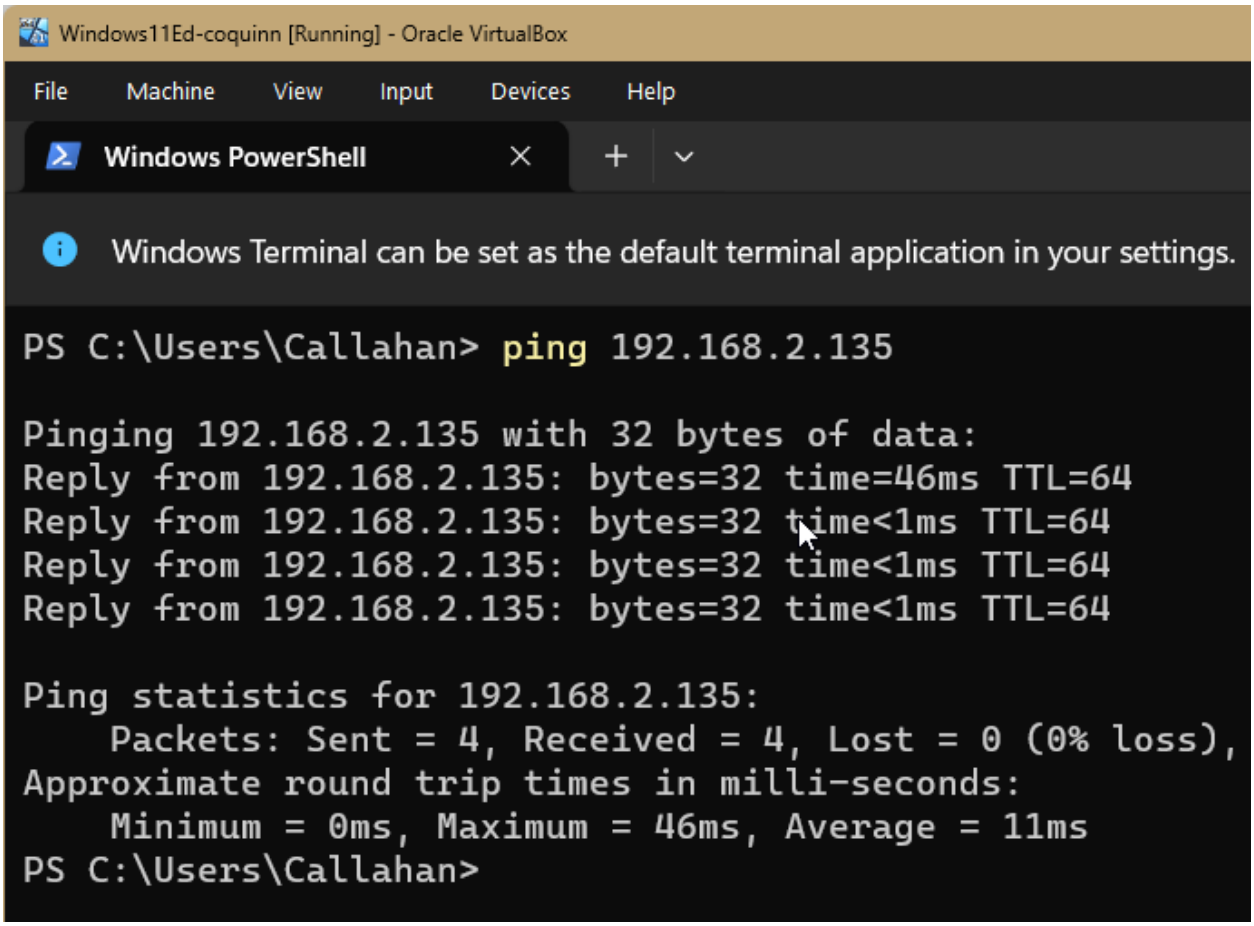
- What changes after enabling the firewall?
  - In my case, the messages still went through. I checked the rules and it said all incoming traffic should be blocked. So I'm not sure if it was because the IP of the sender was on the same segment, or if something did not work properly.
- How does this simulate placing a firewall between network segments?
  - Segmenting a network can be done in many ways, including using a firewall. Essentially you block certain traffic/ports/IPs etc. from external segments to ensure if one segment is breached, it cannot easily spread to the next.

3 Allow Limited Access Between Segments

3.1 Allow SSH Access



3.2 PINGING FROM THE WINDOWS VM TO THE UBUNTU VM



3.3 CONNECTING TO THE UBUNTU VM FROM THE WINDOWS VM WITH SSH

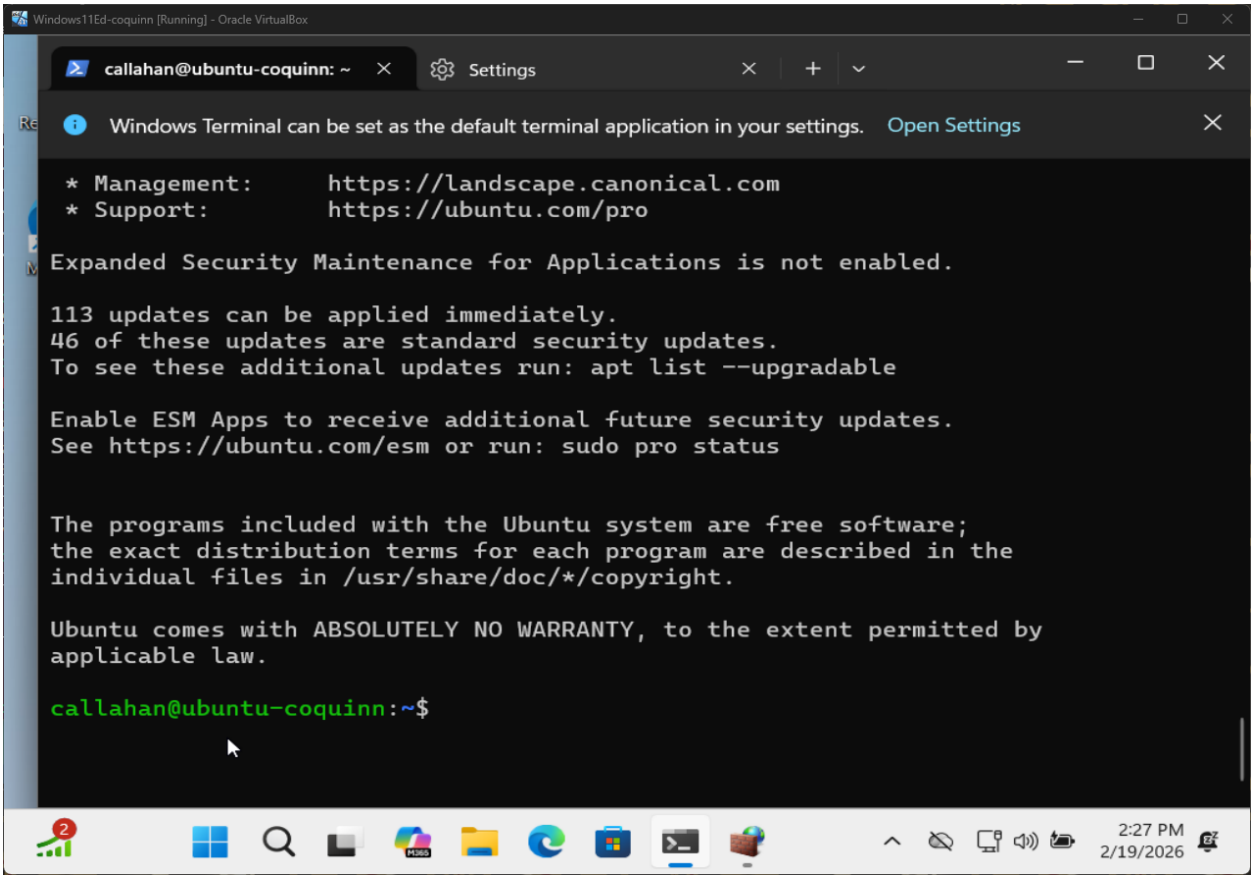
Windows11Ed-coquinn [Running] - Oracle VirtualBox

callahan@ubuntu-coquinn: ~ | Settings

Windows Terminal can be set as the default terminal application in your settings. Open Settings

```
PS C:\Users\Callahan>
PS C:\Users\Callahan> ssh callahan@192.168.2.135
The authenticity of host '192.168.2.135 (192.168.2.135)' can't be established.
ED25519 key fingerprint is SHA256:kVUyx/GQguglcRjrja96XosXYX4B/CFDyRt5txuN9W0.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.2.135' (ED25519) to the list of known hosts
.
callahan@192.168.2.135's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-37-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

113 updates can be applied immediately.
46 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
```

2:26 PM
2/19/2026

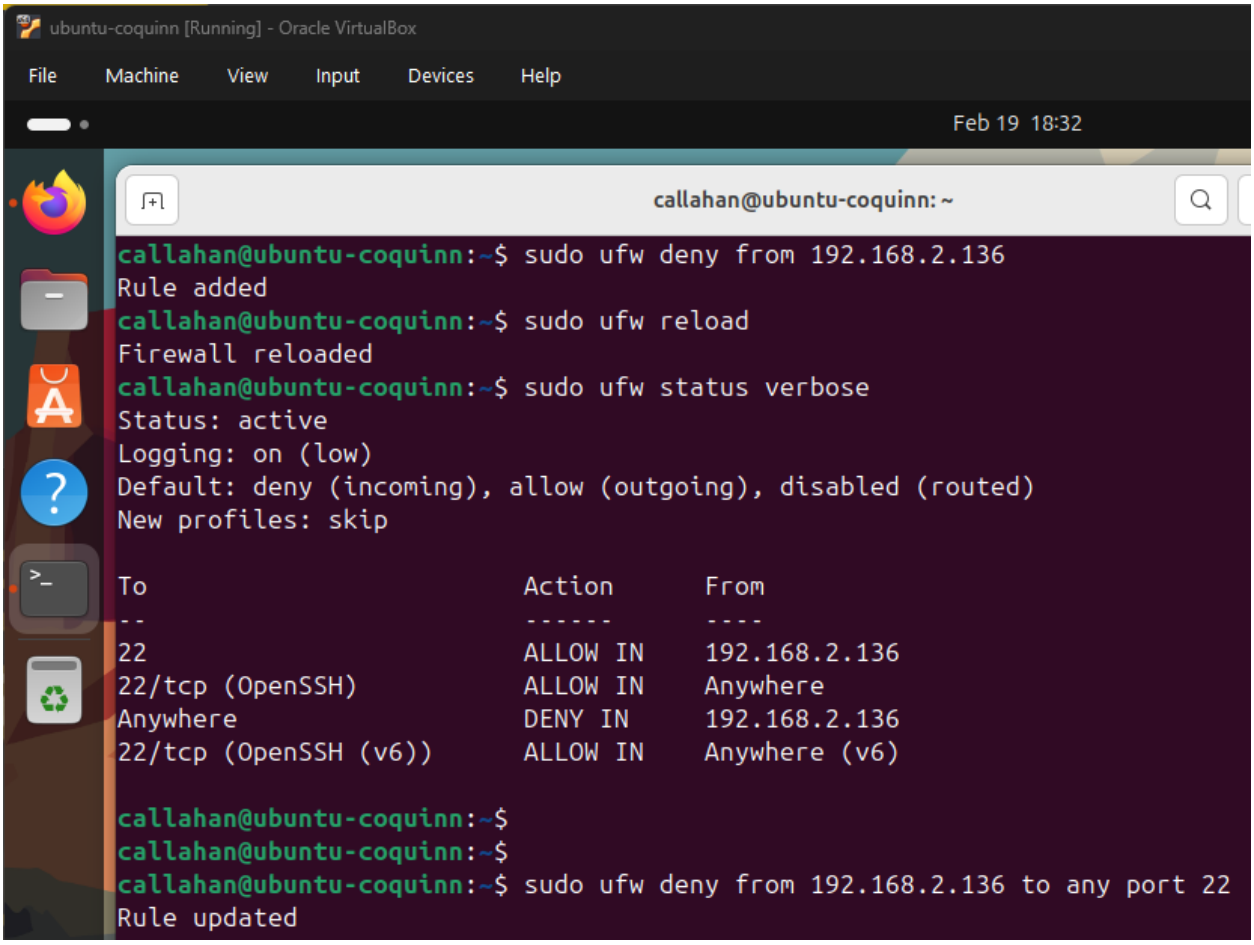


3.4 QUESTIONS

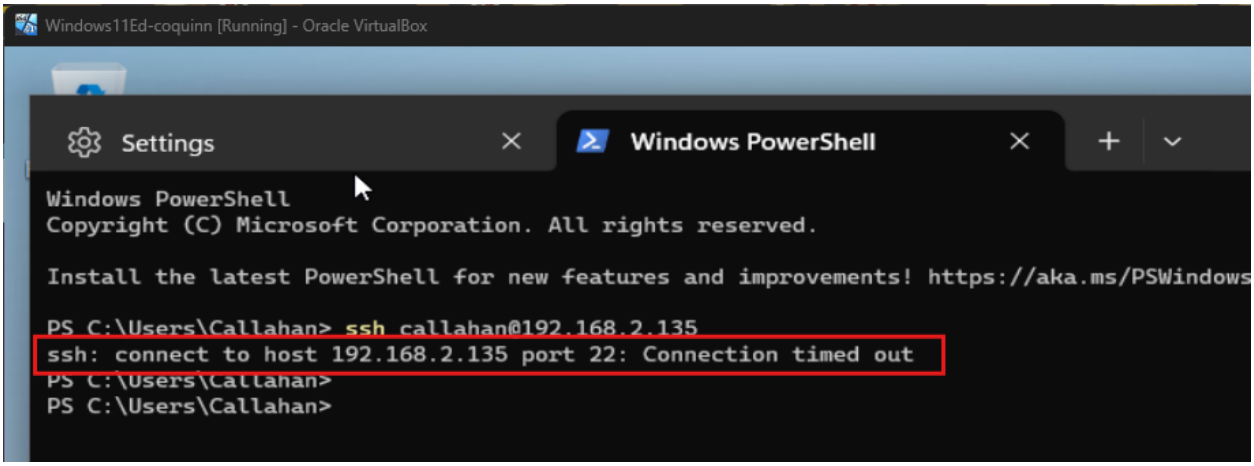
- Why should only required services be allowed between network segments?
  - The less services that are running/enabled/open, the smaller the attack surface is between segments.
- How does this reduce risk during an incident?
  - It is an effective mitigation tactic because it closes more doors an attacker can use to jump from one segment to another. It is super important to take note of open ports and running services, as every opening creates a potential for later movement.

4 Simulating a Network Quarantine

4.1 Block all traffic from the windows system



4.2 Trying to connect from the windows VM again



4.3 QUESTIONS

- How does this simulate a quarantine VLAN?
  - It simulates a VLAN quarantine by blocking all traffic from an exposed device/segment.
- Why should infected systems remain powered on during an incident?
  - Infected devices should remain powered on because depending on the attack, powering off may cause the virus to cause more harm (system corruption, data loss) or trigger the virus to spread laterally. Also, because of these effects, it may significantly hinder the investigation process.
- How does containment reduce ransomware spread?
  - It can stop ransomware from spreading to other network segments by blocking all traffic. This can potentially save more critical systems (ex. servers) or more critical data from being compromised.
- Why is segmentation critical for incident response?
  - Segmenting is critical to both reduce further damage and spread, as well as preserving important evidence for investigating the compromise.